CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 7



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Recap of Previous Lecture



Topic Direct Mapping

Topic Set Associative Mapping

Topic Fully Associative Mapping

Topics to be Covered

5) 17 (3
15
2

17 % 5 = 2



Topic Block Replacement

Topic Miss Penalty



cm ⇒ 64 blocks



256 blocks

[Question]

#Q. Cache size = 256KB

MM address = 32 bits

Block size = 32 Bytes $= 2^5$

no. of blocks in cm $= \frac{256\ KB}{32\ B} = \frac{2^{18}}{2^5} = 2^{13}$

Tag and Tag directory for direct, 4-way set associative and fully associative

Direct



Tag directory size

$= 2^{13} * 14$ bits

4-way

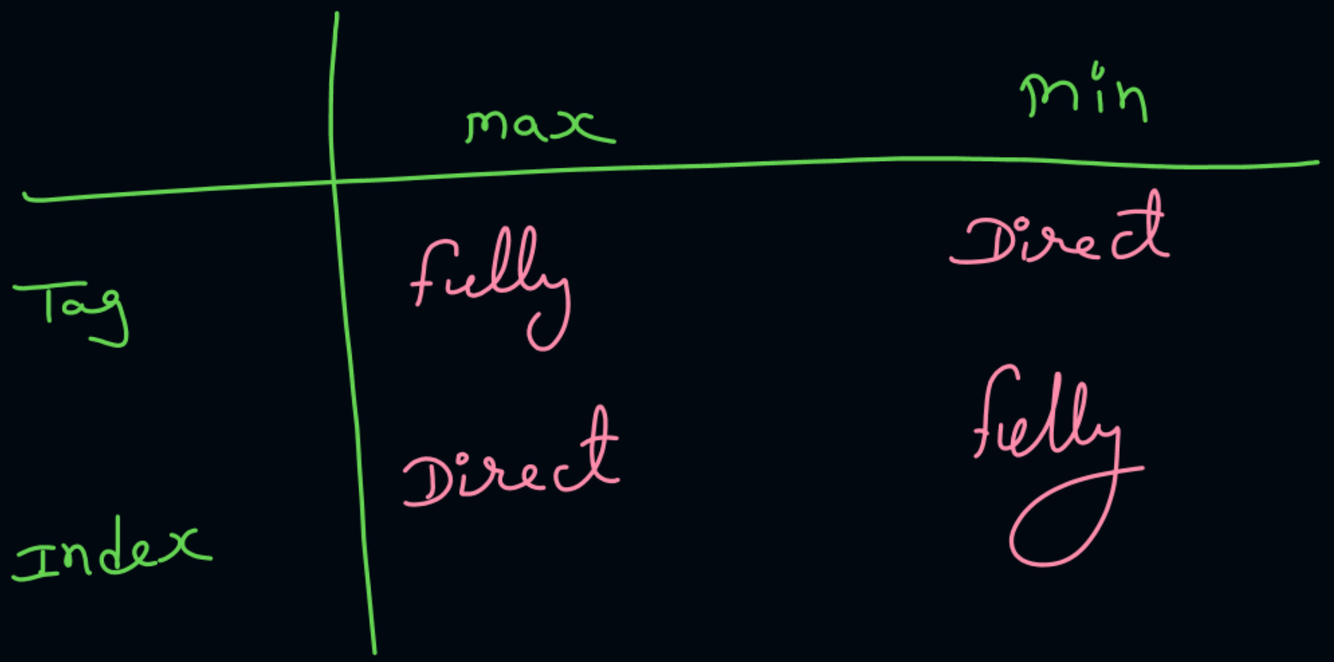


$2^{13} * 16$ bits

fully



$2^{13} * 27$ bits


max
min
Tag
fully
Direct
Index
Direct
fully

Ques) fully ass. cache

no. of blocks in cache = 2^{12}

no. of blocks in mm = 2^{24} ⇒ mm block no = 24 bits

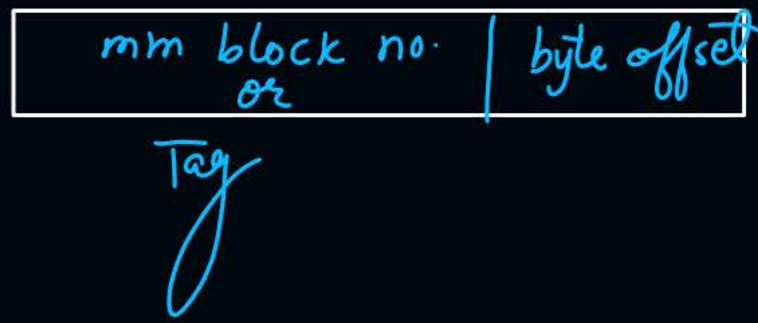
Tag directory size = 96 k bits ?

= 2^{12} * 24 bits

= 96 k bits

fully :-

mm add.



Ques) Direct mapping

cm size = 1KB

block size = 16 B

mm add. = 18 bits

mm add 1. $(5672)_{10}$

2. $(8293)_{10}$

Tag = $(5)_{10}$

cm block no. = $(34)_{10}$

byte = $(8)_{10}$

no. of blocks in cache = $\frac{1KB}{16B} = 64$

mm block no. = $\left\lfloor \frac{5672}{16} \right\rfloor = 354$

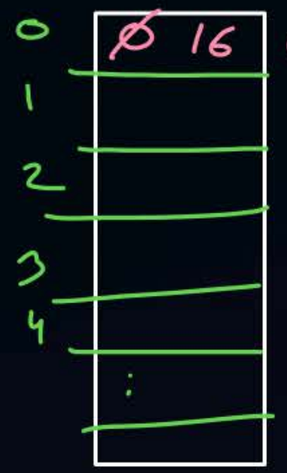
byte offset = 5672 % 16 = 8

Tag = $\left\lfloor \frac{354}{64} \right\rfloor = 5$

cm block no. = 354 % 64 = 34

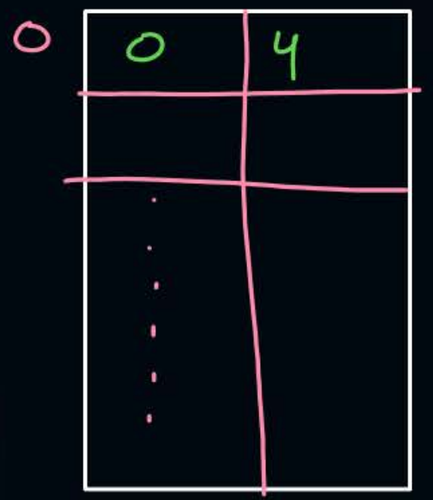
Topic : Block Replacement

Direct



No replacement policy is needed.

2-way set ass.



now CPU Req. for block 8

⇓

block 8 will be coming to set 0, but here which block to be replaced.

⇓

block replacement policy to be used.

Fully



Replacement policies :-

1. FIFO (First In First out)
2. LRU (Least Recently Used) ⇒ The block which has not been used since longest period of time, is replaced.
3. optimal

[MCQ]

#Q. Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is:

8, 12, 0, 12, 8

no. of sets = $\frac{4}{2} = 2$

8%2 = 0

12%2 = 0

0%2 = 0



A 2

B 3

C 4

D 5

[MCQ]

cm block no = (mm block no) % 8

#Q. Consider a Direct Mapped Cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order

3, 5, 2, 8, 0, 63, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24.

Which of the following memory blocks will not be in the cache at the end of the sequence?

A 3

B ✓ 18

C 20

D 30

0	~~8~~ ~~0~~ ~~16~~ 24
1	~~9~~ ~~17~~ ~~25~~ 17
2	~~2~~ 18 ~~2~~ 82
3	3
4	20
5	5
6	30
7	63

3 % 8 = 3

no. of hits = 3

no. of miss = 17

hit ratio = $\frac{3}{20}$

miss ratio = $\frac{17}{20}$

[MCQ]

H.W.

#Q. Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 block and the request for memory blocks is in the following order:

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155

Which one of the following memory block will not be in cache if LRU replacement policy is used?

A 3

B 8

C 129

D 216

[MCQ]

#Q. Consider a fully associative cache with 8 cache blocks (numbered 0–7) and the following sequence of memory block requests:

4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7

If LRU replacement policy is used, which cache block will have memory block 7?

	0	1	2	3	4	5	6	7
0	4 45	3 22	25	8	~~19~~ 3	~~6~~ 7	16	35

A 4

B 5 ✓

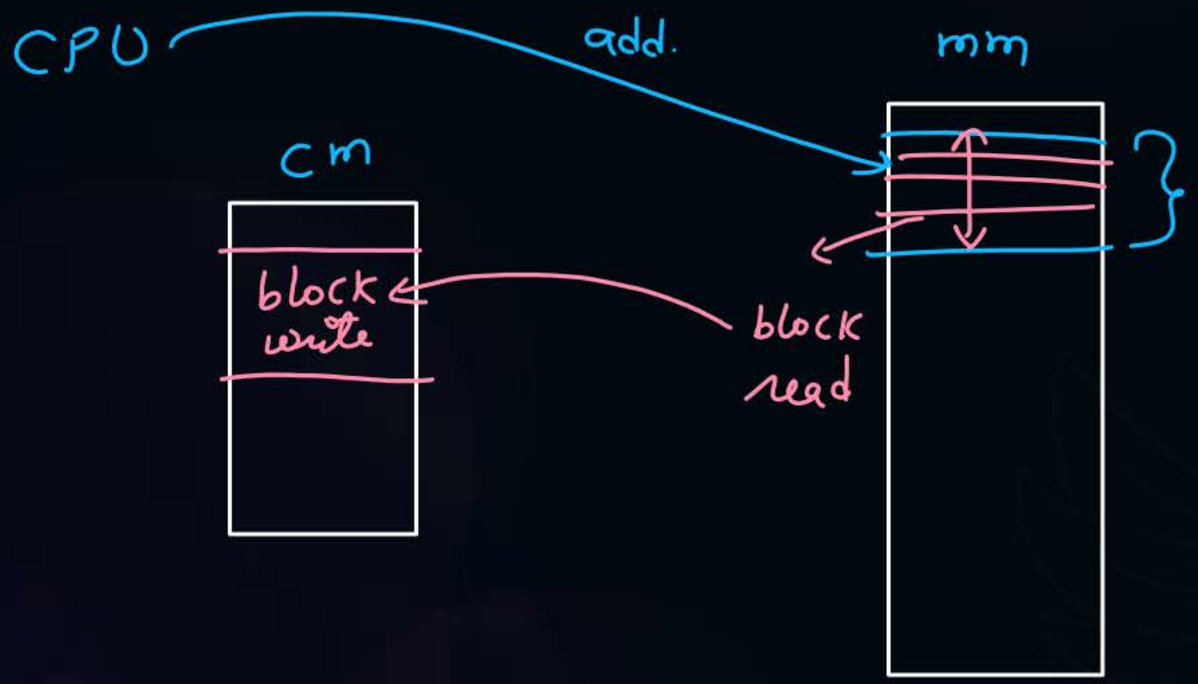
C 6

D 7



Topic : Cache Miss Penalty

Time required to bring a missed block from main memory to cache





Topic : Cache Miss Penalty

Assume:

Cycles required to send address to memory : 1 cycle

Cycles required to access 1 main memory cell : 10 cycles

Cycles required to transfer 1 cell data to cache : 1 cycle

only once



Cache Block Size	Main memory cell size	Miss Penalty
4 bytes	1 byte	$1+(4*10)+(4*1)=45$ cycles = 22.5 ns
4 bytes	2 bytes	$1+(2*10)+(2*1)=23$ cycles = 11.5 ns
4 bytes	4 bytes	$1+(1*10)+(1*1)=12$ cycles = 6 ns

If CPU clock rate = 2GHz then | 1 cycle time = $\frac{1}{2GHz}$ = 0.5 ns

[NAT]

GATE-PYQ

#Q. A certain processor deploys a single-level cache. The cache block size is 8 words, and the word size is 4 bytes. The memory system uses a 60 MHz clock. To service a cache-miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is __160__ $\times 10^6$ bytes/sec?

Miss penalty = 1 + 3 + 8 = 12 cycles = $\frac{12}{60\text{ MHz}} = \frac{1}{5}$ μsec = 0.2 μsec

In 0.2 μsec, data accessed = 8 words = 8 * 4 = 32 B

In 1 sec, —— || —— = $\frac{32\ B}{0.2 * 10^{-6}\ sec}$

$= 160 * 10^{6}$ B/sec



Topic : Types of Cache Misses

1. Cold or Compulsory Miss
2. Capacity Miss
3. Conflict Miss

} → for repeated access of a block



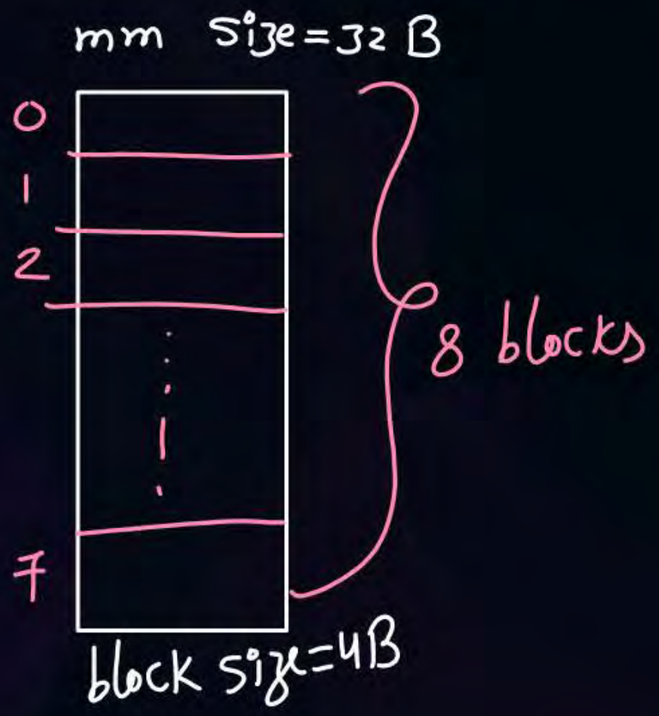
Topic : Types of Cache Misses

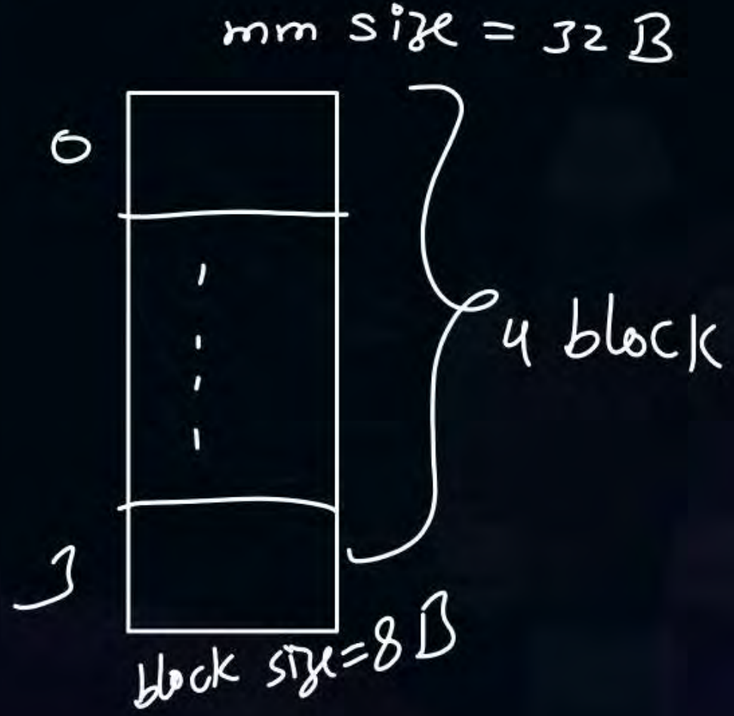
1. Cold or Compulsory Miss

First time access of a block will always cause a miss

To reduce Cold misses: Increase block size

max
Cold miss = 8





max
cold
misses = 4



Topic : Types of Cache Misses

2. Capacity Miss

If It is not cold miss

If cache is full and hence miss occurs.

To reduce Capacity misses: Increase cache size

Topic : Types of Cache Misses

3. Conflict Miss

If it is not cold and capacity.

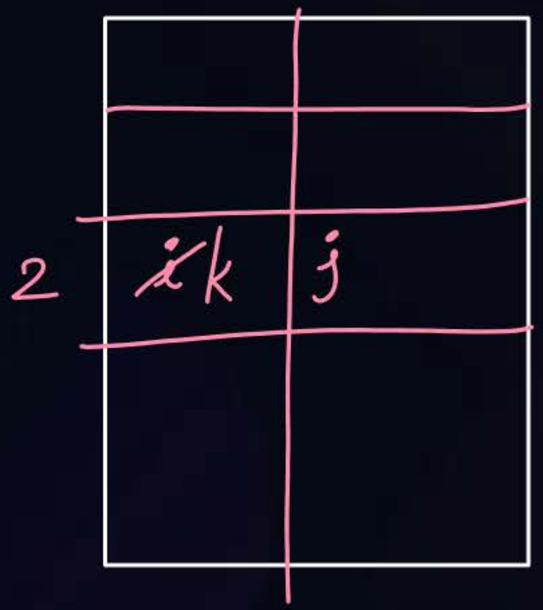
If cache set is full and hence miss occurs.

To reduce Conflict misses: Increase associative

In fully ass. cache

no. of conflict miss = 0





block i requested by CPU

⇓

conflict miss

because if the only set in cache is full then it is capacity miss.



Topic : Example

- Cache is empty Initially
- No. of blocks in cache = 4
- 2-way set associative cache
- LRU replacement policy
- CPU requests for main memory blocks

no. of sets = $\frac{4}{2} = 2$

Cold Cold Cold Conflict Cold Cold Cold Capacity

0, 4, 0, 8, 0, 4, 1, 3, 1, 5, 1, 3

0	0	~~4~~ ~~8~~ 4
1	1	~~3~~ ~~5~~ 3



2 mins Summary

Topic Block Replacement

Topic Miss Penalty

H.W.

GATE – 2017

No. of conflict miss = ?



Happy Learning

THANK - YOU